
MA 529: Statistical Inference

Course Code	MA 529
Course Name	Statistical Inference
L-T-P-C	3-1-0-4
Prerequisites	MA-524 (Probability and Statistics)
Intended for	M.Sc./M.S./PhD/ B.Tech
Mutual Exclusion	NA
Approval	50th BoA (Jul 2023)

Course Contents

- **Module 1:** Random sample, Statistics, Order statistics, Sampling distributions, Parametric point estimation, Estimator, Unbiasedness, Sufficiency, Minimal sufficiency, Factorization theorem, Rao-Blackwell theorem, Completeness, Lehmann-Scheffe theorem, UMVUE, Basu's Theorem, Lower bounds for the variance of an estimator, Frechet-Rao-Cramer, Bhattacharya, Chapman- Robbins-Keifer inequalities. (16 Hours)
- **Module 2:** Consistency, Efficiency, Method of moments and method of maximum likelihood, Bayes estimators and Minimax Procedure, Invariance, Best equivariant estimators. (10 Hours)
- **Module 3:** Tests of hypothesis, Simple and composite hypothesis, Types of error, Neyman-Pearson Lemma, Families with monotone likelihood ratio, UMP, UMP unbiased and UMP invariant tests, Likelihood ratio tests- applications to one sample and two sample problems, Chi-square tests, Bayes tests, Methods for finding confidence intervals, shortest length confidence intervals, Bayesian confidence interval. (16 Hours)

Text Books

1. Main Text Book: Statistical Inference, George Casella and Roger L. Berger, Duxbury Press, second edition 2001.
2. An Introduction to Probability and Statistics, Vijay K Rohatgi and A. K. Md. Ehsanes Saleh, John Wiley, second edition, 2001.

References

1. A. M. Mood, F. A. Graybill and D. C. Boes, **Introduction to the theory of Statistics**, 3rd edition, McGraw Hill Education, 2017.
2. J. Shao, **Mathematical Statistics**, Springer, 1998.
3. E. L. Lehmann, G. Casella, **Theory of Point Estimation**, Springer, 2006.
4. E. L. Lehmann, J. P. Romano, **Testing of Statistical Hypothesis**, Springer, 2006.